Inverse Relations and Functions

Name _____

Algebra 2 Lesson 5-6 (fast track) Date _____ Period _____

Learning goal: I can find the inverse of a function from a table, an equation, or a graph, and restrict the domain so that the inverse is a function.

Language goal: I can explain how a function and its inverse are related using the words *inverse*, *reflection*, *domain*, *range*, and *restrict*.

Spanish: inverse function = *función inversa*; inverse relation = *relación inversa*; relation = *relación*.

On the quiz: Example 2 gets you (2) sketching a line and its inverse and (5, bonus) which line is the inverse of $f(x) = 2x + 4$. Example 3 gets you (3) restricting the domain of a parabola. Full Savvas quiz only: (1) the table of an inverse ("The idea") and (4) a cube-root inverse (same switch and solve).

The idea: the inverse exchanges the roles of the domain and range.

Tables. Switch the rows (or columns).

x	0	1	2	3	4
$f(x)$	2	4	6	8	10

x	2	4	6	8	10
$f^{-1}(x)$	0	1	2	3	4

Algebra. Switch x and y, then solve for y.

$y = x + 9$
$x = y + 9$
$y = x - 9$

Graphs. Reflect the graph across the line $y = x$: a point (x, y) on f becomes (y, x) on the inverse.

Example 2: Find an Equation of an Inverse Relation

Let $f(x) = x^2$.

A. How can you represent the inverse relation of f algebraically?

1. Write y for $f(x)$: $y = x^2$

2. Switch the roles of x and y: $x = y^2$

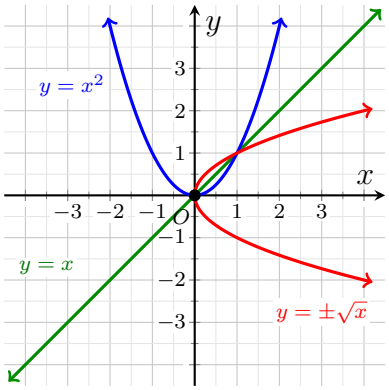
3. Solve for y: $y = \pm\sqrt{x}$

The inverse of f is $y = \pm\sqrt{x}$. Check: $(2, 4)$ is on f; $(4, 2)$ and $(4, -2)$ are on the inverse.

B. How are the graphs of $y = x^2$ and $y = \pm\sqrt{x}$ related?

The graph of the inverse is the **reflection** of the graph of $y = x^2$ across the line $y = x$.

- The **range** of f is $y \geq 0$, so the **domain** of the inverse is $x \geq 0$.
- The inverse is **not** a function: $x = 4$ gives two values, $y = 2$ and $y = -2$.



Say it: The inverse of $f(x) = x^2$ is $y = \pm\sqrt{x}$. Its graph is the _____ of the graph of f across the line _____. The range of f becomes the _____ of the inverse. The inverse is not a function because _____

Try It 2: Find the inverse of a line

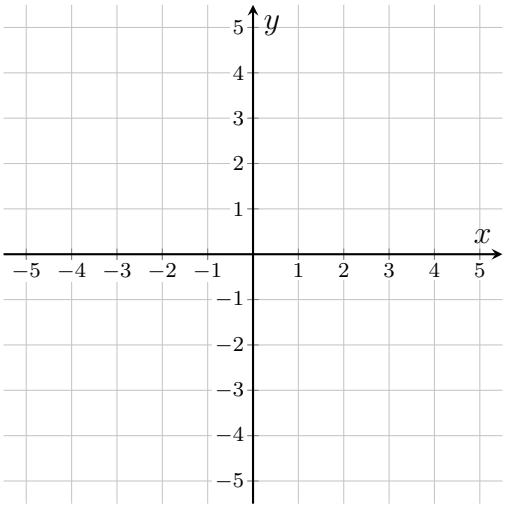
Let $f(x) = 2x + 1$.

a. Write an equation to represent the inverse of f.
$y = 2x + 1 \quad \rightarrow \quad x =$ _____

_____

_____

Inverse: $y =$ _____



b. How can you use the graph of f to determine if the inverse of f is a function? Explain your answer. Hint: sketch f, the line $y = x$, and the reflection of f on the grid.

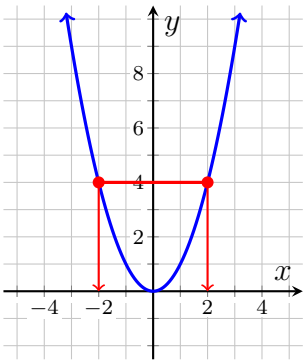
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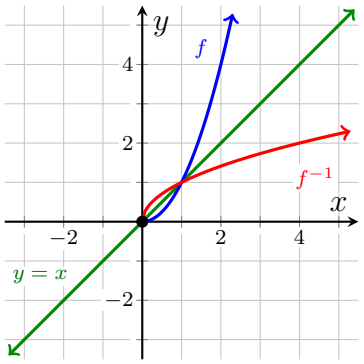
Example 3: Restrict a Domain to Produce an Inverse Function

Consider again the function $f(x) = x^2$. Under what domain will the inverse relation be a function?



Two x-values, one y-value.

1. $(-2, 4)$ and $(2, 4)$ are both on f. The inverse would contain $(4, -2)$ and $(4, 2)$: two outputs for $x = 4$, so it is **not** a function.
2. For the inverse to be a function, all the duplicate y-values must be eliminated. **Restrict** the domain of f to $x \geq 0$ (keep the right half).
3. Switch and solve: $x = y^2$, $y = \sqrt{x}$ (the + root: outputs of f^{-1} are inputs of f, so $y \geq 0$).
4. $f^{-1}(x) = \sqrt{x}$, with domain $x \geq 0$ (the range of f).



$f(x) = x^2$, $x \geq 0$, and $f^{-1}(x) = \sqrt{x}$.

Rule: if any horizontal line crosses the graph of f more than once, the inverse is not a function. Another restriction works too: $x \leq 0$ gives $f^{-1}(x) = -\sqrt{x}$. For any parabola, restrict to one side of the vertex.

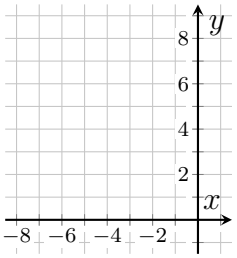
Say it: To make the inverse a function, I _____ the domain of f to _____, so that no horizontal line crosses the graph twice. Then $f^{-1}(x) =$ _____, and its domain is the _____ of f.

Try It 3: Restrict the domain, then find the inverse

Find the inverse of each function by identifying an appropriate restriction of its domain.

a. $f(x) = x^2 + 8x + 16$

$f(x) = (_____)^2$, vertex _____



Restrict the domain of f to _____

Switch and solve:

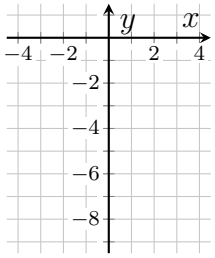
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$f^{-1}(x) =$ _____, domain _____

b. $f(x) = x^2 - 9$

Vertex _____



Restrict the domain of f to _____

Switch and solve:

_____

_____

$f^{-1}(x) =$ _____, domain _____

Say it: I _____ the domain because, without the restriction, a horizontal line crosses the graph of f _____ times, so the inverse would not be a _____.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
ZRQ ★	Find inverse functions and relations	Quiz 3 and Quiz 5 (bonus): switch and solve (Examples 2 and 3); also Quiz 4 (full Savvas quiz only)
K2W ★	Graphs of inverse functions	Quiz 2: reflect a line across $y = x$ (Example 2B)
9KT ★	Identify inverse functions	Quiz 5 (bonus): which function is the inverse
YLX	Find values of inverse functions from tables	Optional: Quiz 1 (full Savvas quiz only), switch x and y in a table
LSG	Graph a linear function	Warm-up for Quiz 2: graph a line
UAJ	Graph square root functions	Warm-up for Quiz 3: the graph of $y = \sqrt{x}$ (Example 3)

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Which powers have inverse functions? (DOK 3)

The table shows several functions and some of the inverses of those functions. The table also shows whether some of the inverses are functions.

Function	**Inverse**	**Is the inverse a function?**
$f(x) = x$	$f^{-1}(x) = x$	yes
$g(x) = x^2$	$g^{-1}(x) = \pm\sqrt{x}$	no
$h(x) = x^3$	$h^{-1}(x) = \sqrt[3]{x}$	yes
$k(x) = x^4$		
$m(x) = x^5$		
$n(x) = x^6$		

Part A Determine the inverses of the remaining functions in the table.
Part B Determine if the inverses of the remaining functions in the table are functions.
Part C Make a conjecture about the power of a function if the inverse of that function is a function.

Before the quiz, finish these sentences.

To find an inverse, I switch the _____ and the _____.

The graph of the inverse is the _____ of the graph of f across the line _____.

If a horizontal line crosses the graph of f twice, I _____ the domain so that the inverse is a function.